TEXORA · LECTURE SHEET

The Blowroom

The blowroom opens baled cotton into small tufts, removes most of its trash, blends the material and delivers it as a uniform lap. This sheet treats the blowroom as a calculation problem: cleaning efficiency, waste percentage, lap weight and production rate are the numbers that decide whether the line is doing its job.

YM-3207 · ADVANCED SPINNING CALCULATIONS · TOPIC 1 · SPECIALIZATION — YARN MANUFACTURING

1. Objects of the blowroom

Raw cotton arrives compressed into bales of about 170 kg, containing leaf, seed fragments, dust and sand picked up in the field and the gin. Before any fine machine can work on it, four things must happen. **Opening** reduces the dense bale material to tufts small enough for trash to fall out of them. **Cleaning** removes the liberated trash at grid bars, perforated cages and suction points. **Blending** mixes material from many bales so that variations in staple, micronaire and colour average out. **Lap formation** (or, in modern lines, chute feeding) delivers the opened material forward at a controlled, uniform weight per unit length. Every later process inherits the blowroom's work: trash left here becomes neps and ends down later, and a lap that varies in weight reappears as count variation in the yarn.

The governing principle is *gentle, progressive opening*. Material should meet mild action many times rather than violent action once, because a beater that tears tufts apart also breaks fibres and rolls soft trash into neps that no later machine can remove. The calculation side of the blowroom is therefore a balance: cleaning efficiency high enough, fibre damage and good-fibre loss low enough.

2. The machine sequence



Figure 1 — A conventional blowroom line. Opening intensity rises step by step; cleaning happens wherever opened tufts pass over an extraction point.

Figure 1 — Blowroom sequence: bale plucker, pre-opener, mixing, fine opener and scutcher, with waste extraction (red) at each opening point.

At each beater the tufts are struck against grid bars; centrifugal action throws the heavy trash through the gaps while the light fibre is carried on by the air stream. The proportion of trash removed at a point depends on tuft size (smaller tufts release trash more easily), beater speed, and the grid-bar setting — the distance and angle between beater and grid. Wider grid settings and higher speeds clean more but also throw out more good fibre, which is why cleaning is always quoted together with waste.

3. Cleaning efficiency

Cleaning efficiency compares the trash actually removed with the trash that entered. It must be calculated on **trash masses**, not on percentages alone, because the total mass also falls as waste is extracted.

$$\text{Cleaning efficiency (\%)} = (\text{trash in feed} - \text{trash in delivery}) / \text{trash in feed} \times 100$$

WORKED EXAMPLE — CLEANING EFFICIENCY OF A BLOWROOM LINE

A line processes 100 kg of cotton containing 4.8% trash. It delivers 92 kg of lap (total waste 8%) and the lap contains 1.5% trash.

Trash in feed = 4.8% of 100 kg = 4.80 kg.

Trash in lap = 1.5% of 92 kg = 1.38 kg.

Trash removed = 4.80 − 1.38 = 3.42 kg.

Cleaning efficiency = 3.42 / 4.80 × 100 = **71.3%**.

The remaining 4.58 kg of the 8 kg waste is good fibre and fly lost with the droppings — the price paid for cleaning. A typical well-set blowroom reaches 60–75% cleaning efficiency on medium cottons; pushing far beyond that usually costs more in fibre loss and nep generation than it gains.

Two cautions belong with this calculation. First, trash percentage in the lap is measured on a small sample and varies; the test should be repeated. Second, cleaning efficiency says nothing about *where*

the remaining trash sits: fine dust embedded in the tuft survives to the card, which is designed to finish the job.

4. Waste percentage and its composition

$$\text{Blowroom waste (\%)} = (\text{mass fed} - \text{mass of lap delivered}) / \text{mass fed} \times 100$$

In the example above the waste is (100 − 92)/100 = **8.0%**. Waste is not one material. *Droppings* under the beaters contain most of the trash plus some fibre; *filter and cage waste* is mostly short fibre and dust; a small part is *invisible loss* — moisture and dust that leave with the exhaust air and are found only by difference. The usable part of blowroom droppings can be reprocessed for coarse counts, but for fine counts it is sold off; either way it is not in the lap, so it counts as waste for this department.

EXTRACTION POINT	MAIN MATERIAL REMOVED	TYPICAL SHARE OF FEED
Bale plucker / pre-opener grids	Heavy trash, seed, leaf	2–3%
Fine opener grids	Trash, broken seed, fibre clusters	2–3%
Scutcher cage and grids	Dust, short fibre, fine trash	1–2%
Filters and invisible loss	Dust, moisture	1–2%

5. Lap weight and production

The scutcher delivers the lap at a set weight per metre, controlled by the pedal or electronic regulator that varies feed as the incoming sheet thickness changes. Lap weight is checked by weighing a full lap of known length; variation between laps and within a lap is expressed as CV% and is the first evenness figure of the whole mill.

$$\text{Production (kg/h)} = \text{delivery speed (m/min)} \times \text{lap weight (kg/m)} \times 60$$

WORKED EXAMPLE — BLOWROOM PRODUCTION AND CARDS SERVED

A scutcher delivers lap at 11 m/min and the lap weighs 450 g/m = 0.450 kg/m.

Production = 11 × 0.450 × 60 = **297 kg/h**.

A card fed with this lap delivers about 15.3 kg/h of sliver and loses roughly 5% at carding, so each card consumes about 15.3 / 0.95 = 16.1 kg/h of lap.

Cards served by one line = 297 / 16.1 ≈ **18 cards**.

This is why blowroom output is planned in lines matched to card groups, not machine by machine: one line at this rate balances a group of about eighteen cards.

Lap length is chosen to suit handling and card creel practice (a common lap is about 40 m, weighing 0.450 × 40 = 18 kg). When a mill changes to chute feed, the same production calculation applies with the card's consumption rate in place of lap weight and delivery speed.

6. Faults that begin in the blowroom

- **High nep count in the lap** — beaters too fast or grids too close for the cotton, or tufts too large reaching a fast beater; neps formed here survive carding in part and show as yarn imperfections.
- **Stringy, unopened lap** — overloading the line (production rate above the openers' capacity) so tufts pass through unopened.
- **Excessive waste** — grid settings too open or beater-to-grid distance too large; good fibre leaves with the trash and cleaning efficiency rises while yield falls.
- **Lap weight variation** — irregular feed to the scutcher or a faulty regulator; appears directly as card sliver variation.

Glossary

Tuft — a small cluster of fibres, the unit in which opened material travels. **Grid bars** — the triangular-section bars under a beater through which trash escapes. **Cleaning efficiency** — percentage of incoming trash removed. **Lap** — the rolled sheet of opened cotton delivered by the scutcher. **Droppings** — the waste collected under beaters. **Invisible loss** — waste found only as the difference between total input and all weighed outputs. **Scutcher** — the final blowroom machine, which forms the lap.

Key points

• The blowroom opens, cleans, blends and delivers; opening must be progressive and gentle.
• Cleaning efficiency is calculated on trash masses: CE = (trash in − trash out)/trash in × 100; about 71% in the worked example.
• Total waste (8% in the example) always exceeds trash removed — the difference is fibre and invisible loss.
• Production = delivery speed × lap weight × 60; the example line makes 297 kg/h and feeds about 18 cards.
• Blowroom faults — neps, unopened tufts, lap variation — cannot be fully repaired downstream.

Practice

1. A blowroom receives 250 kg of cotton with 5.6% trash and delivers 232 kg of lap containing 1.8% trash. Calculate the cleaning efficiency and the total waste percentage.
2. A scutcher delivers at 12 m/min with a lap of 420 g/m. Calculate the production in kg/h, and the weight of a 38 m lap.
3. Explain why raising beater speed can raise cleaning efficiency yet still be bad practice. Name the two penalties.

Answer key

1. Trash in = 5.6% of 250 = 14.0 kg. Trash in lap = 1.8% of 232 = 4.176 kg. Removed = 14.0 − 4.176 = 9.824 kg. CE = 9.824/14.0 × 100 = **70.2%**. Waste = (250 − 232)/250 × 100 = **7.2%**.
2. Production = 12 × 0.420 × 60 = **302.4 kg/h**. Lap weight = 0.420 × 38 = **15.96 kg**.
3. Higher beater speed strikes tufts harder and throws more trash through the grids. The penalties are (1) fibre rupture — shorter fibres, lower yarn strength — and (2) nep formation, as soft trash and fibre clusters are rolled into neps; both also tend to raise good-fibre loss in the droppings.
